



Software Development Projects

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Main Goal:

This roadmap describes three software projects I plan to build as I develop my general software engineering skills. Each project will introduce a different kind of programming challenge: first building a useful command-line program, then creating a desktop application that saves data, and finally developing a faster tool that works directly with files and larger amounts of information. Through these projects, I plan to practice problem solving, code organization, testing, debugging, and documenting software from start to finish.

Contents

Projects	Project type	Key take-away
Personal Study Planner	Command-line application	Programming fundamentals and problem solving
Personal Library Manager	Desktop application	Object-oriented design, user interfaces, and data storage

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01 Personal Study Planner

Goal

I plan to build a program that helps students organize courses, assignments, study sessions, and due dates. Users will be able to add and update tasks, mark work as complete, and review what they need to do. The program will start in the command line, so I can focus on how software works before adding a graphical interface.

This will be my first project focused on general software development. I want to strengthen my programming fundamentals by breaking a practical problem into smaller steps, handling user input, and organizing information in a useful way. These skills apply to many kinds of software, including desktop, mobile, web, and command-line applications.

Experience I plan to gain

- Use variables, conditionals, loops, functions, and collections to solve problems
- Organize a program into files and reusable functions
- Validate user input and handle errors without crashing
- Read and write data to a file so information is saved between runs
- Debug issues and test different user scenarios
- Use Git and GitHub to track changes and document a project

Planned technology

Core stack Python · JSON or CSV file storage · Git/GitHub

Tools VS Code · Python debugger · unittest or pytest

First release features

- Add, edit, view, and delete study tasks
- Organize tasks by course
- Save due dates and priority levels
- Mark tasks as complete
- View all tasks or filter by course, date, or completion status
- Save task information locally between program sessions

Build roadmap

- 01** Define the information a study task needs
- 02** Plan the command-line menu and user flows
- 03** Build the task functions using sample data
- 04** Add input checks and helpful error messages
- 05** Save and load tasks from a local file
- 06** Test common and unusual inputs, write a README, and publish the code

02 Personal Library Manager

Goal

I plan to build a desktop application that helps users organize a collection of books, games, or other personal items. Users will be able to add items, search and filter their collection, and keep track of whether an item is owned, borrowed, or finished.

This project will build on the programming fundamentals from the study planner and introduce a graphical user interface and a database. I want to learn how a desktop application responds to user actions, how its different parts should be organized, and how information can be saved reliably. These are useful skills for creating software that people install and use directly on their computers.

Experience I plan to gain

- Use object-oriented programming to represent items and application behavior
- Build a graphical interface with forms, buttons, lists, and search
- Separate the interface, application logic, and data storage
- Store and retrieve structured information with SQLite
- Write tests for important application behavior
- Package an application so another person can install and try it

Planned technology

Core stack Python · PySide6 · SQLite

Tools VS Code · Git/GitHub · pytest · PyInstaller

First release features

- Add, edit, and remove collection items
- Save details such as title, type, creator, and status
- Search and filter the collection
- Store information in a local SQLite database
- Provide clear messages when saving or searching fails
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Build roadmap

- 01** Plan the screens and decide what information each item should store
- 02** Design the database tables and basic application structure
- 03** Build the main window and collection list
- 04** Add forms to create and edit items
- 05** Connect the interface to SQLite
- 06** Add search, filters, and input validation
- 07** Test key actions, package the application, and document how to use it

What I want to be able to show

By completing these projects, I aim to demonstrate growth from programming fundamentals to building reliable applications with user interfaces, saved data, and more complex file-processing behavior. I also want to show how I approach problems: planning the work, organizing code, testing important behavior, debugging issues, and documenting what I learned.

These projects will complement my web development projects by giving me practice with software concepts that apply across different platforms. My plan may change as I learn, but I want each project to become a working piece of software that I can explain and improve.